המרכז
לחינוך סייבר
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Assembly for Reverse Engineering

Registers

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Registers

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- The CPU has some special hardware circuits
- Using them requires zero wait time
- Limited resource
- General purpose registers:
  - Used for all types of calculations
- Special purpose registers
  - IP
  - FLAGS

General Purpose Registers

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- 16 bit assembly:
  - AX – Accumulator register
  - BX – Base address register
  - CX – Count register
  - DX – Data register
  - SI – Source Index
  - DI – Destination Index
  - BP – Base Pointer
  - SP – Stack Pointer

8 Bit Registers

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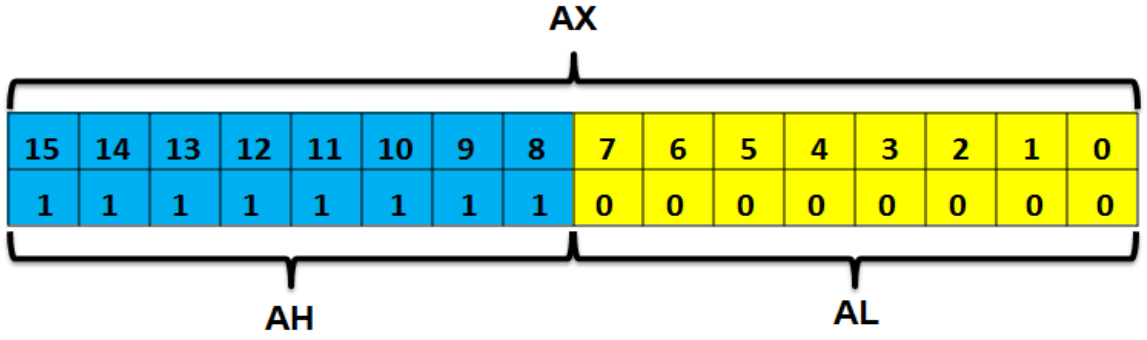
- For byte size operations, use byte registers:

16 bit	8 bit	8 bit
AX	AH	AL
BX	BH	BL
CX	CH	CL
DX	DH	DL

- mov ax, 0xFF00
  - mov ah, 0xFF
  - mov al, 0x00



32 Bit Registers

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- “E” stands for “Extended”

32 Bit	16 Bit	8 Bit
EAX	AX	AL / AH
EBX	BX	BL / BH
ECX	CX	CL / CH
EDX	DX	DL / DH
ESI	SI	
EDI	DI	
EBP	BP	
ESP	SP	

Other General Purpose Regs.

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- ESI / EDI – used for copying data from buffer to another
  - String operations
  - Will show up in some exercises
- EBP / ESP
  - Much of the course will be dedicated (stack, procedures)

EIP -Special Purpose Register

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- Study hands on :-)

```
.text:00402000 ; Attributes: noreturn
.text:00402000
.text:00402000 public start
.text:00402000 start proc near
.text:00402000 mov     eax, 1
.text:00402005 mov     ebx, 2
.text:0040200A add     eax, ebx
.text:0040200C mov     esi, offset aHi ; "hi\r\n"
.text:00402011 call    sub_40309D
.text:00402016 mov     esi, offset aBye ; "bye\r\n"
.text:0040201B call    sub_40309D
.text:00402020 push    0               ; uExitCode
.text:00402022 call    ds:ExitProcess
.text:00402022 start endp
.text:00402022
```

```
EAX 000DFFCC ↳ Stack[0
EBX 002C2000 ↳ TIB[000
ECX 00402000 ↳ start
EDX 00402000 ↳ start
ESI 00402000 ↳ start
EDI 00402000 ↳ start
EBP 000DFF80 ↳ Stack[0
ESP 000DFF74 ↳ Stack[0
EIP 00402000 ↳ start
EFL 00000246
```

Summary

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- We have learned about:
  - CPU's general registers – important for reading code
  - EIP - important for tracing code
- By-products:
  - Practiced number representation
  - Practiced Ollydbg

QUESTIONS